

Hardware Test Automation Engineer — Home Assignment

Part 1 — Automated Testing Strategy

Imagine you join a company that develops electronic products. This company has created a multifunctional I/O device. For an inspiration, we'll use an existing device made by [NI](#), namely [USB-6000](#).

To ensure quality of this product, the company needs to test every product before it's shipped to the customer. The company wants you to take care of automated in-house testing of this product.

Prepare a 3-or-more slide presentation that would outline:

- test strategy for this device (high-level with a few examples of tests);
- architecture of a test system that could implement that test strategy (covering hardware and software aspects of this system).

In case of any doubts or ambiguity, just make an assumption and document it.

Part 2 — Software Development

Imagine Python didn't support imaginary numbers natively but you needed this feature for a project. How would you add this feature? Implement essential functionality, such as addition, subtraction, division, and multiplication. Think about other aspects that need to be covered and either implement them as well or create placeholders with TODOs. Consider adding documentation and some tests.

Work in a local or a GitLab/GitHub based repository, if possible.